

Forbidden valuations: where a halt cannot be

WS3, first execution — an unconditional infinite-time exclusion, and the point where the ladder stops climbing — July 26, 2026

1. Result

On the branch $v_2(x) = v$ the Needle map is affine with fixed point $x^* = 2^v(3 - 2v)/3$, so a run of n consecutive steps of valuation v satisfies $2^{(v+1)n}(3x_n - 2^v(3-2v)) = q_v^n(3x_0 - 2^v(3-2v))$ with $q_v = 2^{v+1} + 3$. If that run ends in a halt, $x_n = 2^m$, and because q_v is odd:

$$q_v^n \text{ divides } 3 \cdot 2^{m-v} - 3 + 2v \quad (*)$$

For $v = 0$ this is $5^n \mid 2^m - 1$, the LTE case already on file ($m \geq 4 \cdot 5^{n-1}$). For general v it is a discrete-logarithm condition, and the new point is that **it is often unsolvable**: $3^{-1}(3-2v)$ need not lie in the subgroup generated by 2 modulo q_v .

Theorem (forbidden valuations). If $(*)$ is unsolvable modulo q_v , then no orbit of the Space Needle can halt at a step whose predecessor has valuation v — at any scale, at any time. Among $v \leq 34$ that excludes

$$v = 1, 4, 7, 9, 10, 12, 13, 18, 19, 21, 22, 23, 24, 25, 26, 28, 31, 32, 33 \quad (19 \text{ of } 35)$$

and since valuation v has frequency $2^{-(v+1)}$, this removes an asymptotic **28.7%** of all steps as possible halt-predecessors. Measured on the actual orbit over its first 200,000 steps: **28.91%** of steps are unconditionally proved not to be followed by a halt. Independent confirmation: brute force over $x < 3,000,000$ finds single-step halting seeds only at valuations $\{0, 2, 3, 6\}$ — every one allowed, none forbidden.

2. The position bound, and why it adds almost nothing

For an allowed v , $(*)$ still pins the halting exponent to one residue class modulo $\text{ord}(2, q_v^n)$, so $m \geq t_{\min}(v, n) + v$. The map expands by at most 3 per step, so $x_N \leq 6 \cdot 3^N$ and $m \leq 1.585N + 2.6$; a halt at the end of a v -run of length n therefore cannot occur before step $N_{\min} = (t_{\min}(v, n) + v - 2.6)/1.585$:

run length n	$v=0$	$v=2$	$v=3$	$v=5$	$v=6$	$v=8$
1	—	4.1	5.9	12	6.6	60
2	—	48	108	2.4e3	1.5e3	1.1e4
3	—	742	2.1e3	5.5e4	1.4e6	2.1e6

The numbers grow fast in n — $t_{\min}$ scales like q_v^{n-1} — but they are racing against m , which grows only *linearly* in the step index. On the real orbit the position bound excludes **8 steps out of 200,000**. Essentially all of the 28.91% comes from the forbidden valuations alone.

That is the honest verdict on this rung of the ladder: it saturates immediately. The exclusion is permanent and unconditional, but it is a fixed 28.7% of steps and it does not improve with more computation, because runs long enough for the position bound to bite are exponentially rare exactly

where the bound would need them to be common.

3. Why the multi-block rungs need Baker, and what changes

The plan proposed grading halting branch words by block count B and climbing the Steiner $\rightarrow$ Simons–de Weger $\rightarrow$ Hercher ladder. Working the algebra out shows why $B = 1$ behaves so differently from $B \geq 2$. Condition (*) constrains m using only the *last* block, because the fixed point x^* does not depend on where the run started. Earlier blocks do not chain into further divisibility conditions on m ; they constrain the value entering the configuration instead. So the congruence method has exactly one rung, and it has now been climbed.

Blocks beyond the last enter only through an inequality. Composing runs of valuations $v_1, \dots, v_B$ and lengths $n_1, \dots, n_B$ and demanding the result be 2^m gives a small linear form in logarithms,

$$| m \log 2 - \sum_i n_i \log(q_{v_i} / 2^{v_i+1}) - \log C | \text{ small,}$$

which is Baker territory proper: Matveev for the lower bound, continued-fraction or LLL reduction to bring the resulting exponent bound down to a finite check. That is a substantially larger piece of work than this one, and the honest expectation from the $B = 1$ experience is that it constrains *which* exponent combinations are possible rather than excluding whole valuations — useful, but a different shape of result. The one encouraging precedent remains April 2026's BB(6) machine settled by Baker–Wüstholz.

4. Verification ledger

claim	status
The run closed form and (*)	PROVED (affine conjugation to the fixed point; q_v odd). Machine-verified at $n = 1$ on random values for $v = 0..3$, and every single-step halting seed below 3,000,000 satisfies (*).
Forbidden valuations for $v \leq 34$	MACHINE-VERIFIED by exact discrete logarithm (baby-step giant-step) modulo q_v . Unsolvability mod q_v implies unsolvability mod q_v^n for all n , so the exclusion holds for runs of every length.
28.91% of the first 200,000 orbit steps excluded	MACHINE-VERIFIED (direct simulation), and consistent with the asymptotic 0.2868 predicted by the valuation frequencies.
The position bound table	MACHINE-VERIFIED. Conservative: run lengths are capped at $n = 3$ when computing $t_{\min}$, which imposes fewer constraints and so can only understate the exclusion.
Non-halting	OPEN. This constrains where a halt can be, not whether one exists; 71% of steps remain unexcluded by this method.

Code: runs.py (algebra, (*), discrete logs, tests), ladder.py (forbidden set, position bounds, orbit exclusion).